

BUSOLINI Hugo
GRANDPERRET Alexi

Ipsa Toulouse
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Numerical simulations of steady, supersonic nozzle flows using Star-CCM+ / CFD TP2

SUMMARY OF RESULTS

The present report covers the three Siemens – Star-CCM+ simulation results and analyses in our converging-diverging Laval nozzle flow modelling project. With an axisymmetric (by revolution) 3D geometry, we will simplify the software implementation of the nozzle by studying a 2D longitudinal section while still conserving coherent flow behavior.

We will firstly model an ideally expanded nozzle to get familiar with the numerical setup and results extraction, before tackling an over-expanded nozzle model and its resulting jet plume in a third and last tutorial.

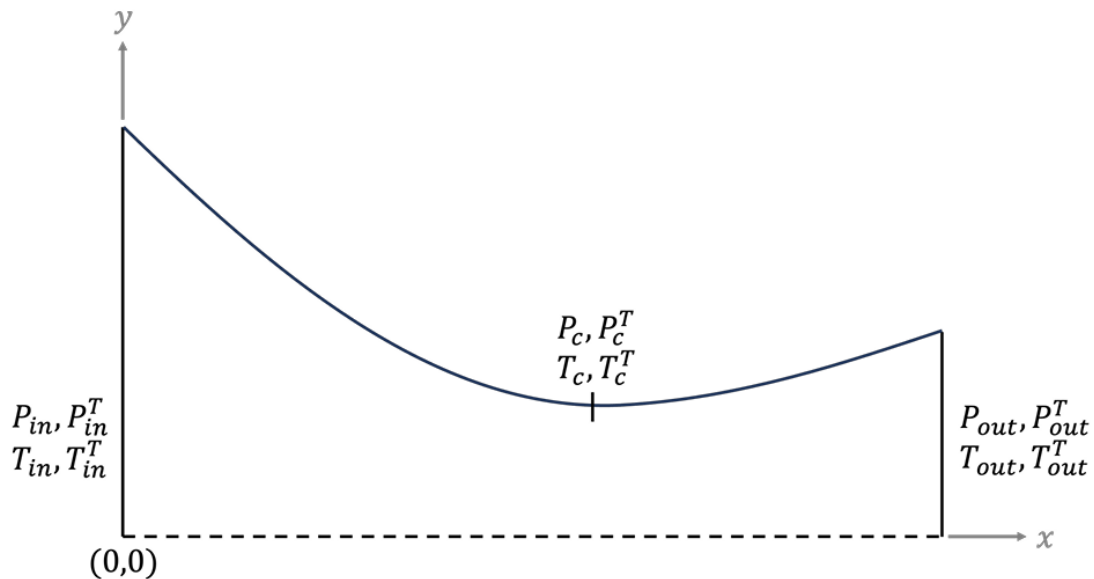


Figure 1 - Sketch of the converging-diverging axisymmetric nozzle

NOTIFICATION

When extracting the throat quantities in subject I, we realized the StarCCM+ spline geometry presented a throat (= minimal section) downstream of the expected position established in the subject. Prior to the second tutorial, we addressed this by imposing a horizontal constraint to the spline at the $x=0.03\text{m}$ throat coordinate, complying with the theoretical model. The numerical $\dot{m}_s$ error dropped from 8.8% (0.518kg/s) to 0.5% (0.568kg/s), validating the physical consistency of this correction.

Having already completed the first subject with a coherent method and physically correct results, we opted to stick with the original assessment indications of a throat at $x=0.03\text{m}$ despite the second tutorial instructions to revert to the original spline (new downstream throat).

Although we understand the “not modelling like other groups” concern, which we conformed to by modifying our geometry for the subject II, we believe that our assessment-aligned simulation and physically correct subject I results remain an equally valid and justified report.

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I – IDEALLY EXPANDED NOZZLE

Question 1 – M_{out} calculation

In an ideal adapted nozzle, no normal shock occurs in the nozzle (between the throat and outlet). Thus, we can calculate the outlet Mach using the section ratio formula from the throat:

$$\frac{A_x}{A_c} = \frac{1}{M_x} \left(\frac{2 + (\gamma - 1)M_x^2}{\gamma + 1} \right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

We will numerically solve this equation, knowing the below sections definition -

$$A_x = A_{x=x_o=0.06} = \pi 0.015^2$$

$$A_c = A_{x=x_c=0.03} = \pi 0.01^2$$

- giving us an outlet Mach of $M_{out} = 2.33$.

Question 2 – P_{out} calculation

Knowing that in an adapted nozzle we have the **conservation of total quantities** (no entropy creation), we can finally use the St-Venant equation to solve for the static outlet pressure and temperature.

$$\begin{cases} P_{s-out} = P_{t-out} \left(1 + \frac{\gamma-1}{2} M_{out}^2 \right)^{-\frac{\gamma}{\gamma-1}} = 10 \left(1 + \frac{1.4-1}{2} 2.33^2 \right)^{-\frac{1.4}{1.4-1}} = 0.765 \Rightarrow P_{s-out} = 0.765 \text{ bar} \\ T_{s-out} = T_{t-out} \left(1 + \frac{\gamma-1}{2} M_{out}^2 \right)^{-1} = 500 \left(1 + \frac{1.4-1}{2} 2.33^2 \right)^{-1} = 239.7 \Rightarrow T_{s-out} = 240 \text{ K} \end{cases}$$

Question 3 – Simulation Initialization

For the initial quantities that will define the first iteration flow of our simulation (to speed up the required time until convergence), *homogeneous conditions are acceptable*.

- $P_{s-ini} = P_{t-ini} = 10 \text{ bar}$
- $T_{s-ini} = T_{t-ini} = 500 \text{ K}$
- $||u_{ini}|| = 0 \text{ m/s}$

We will choose this condition type: **homogeneous conditions**, with a still initial flow ($M=||u||=0$):

Asked in the questions, we justify below the models picked for our simulation.

- The first and most crucial one to match the desired simulation framework is the **axisymmetric** model, implementing in the resolution process the revolution symmetry around the bottom axis of our geometry.

- Secondly, we will stick with the **ideal gas** model to represent the air-fuel mixture travelling in the nozzle. Furthermore, our fluid viscosity is here not represented, hence the **inviscid** model: perfect fluid (therefore with also no heat exchange).
- Concerning the solving itself, we here choose **coupled flow + coupled energy** models as we do not require separate equation-by-equation resolution (=segregated).
- Lastly, simulating a non-time-dependent solution, we impose the **steady** model: the resolution will only approach convergence with no dependencies of the results on time.

Furthermore, we impose a **reference pressure of 0Pa**. Consequently, we adopt a **relative pressure framework**, perfectly acceptable for what we aim to study in these assessments.

Lastly, the two boundary regions are the stagnation inlet region – where total pressure and temperature will be imposed – and the pressure outlet region – where static pressure and temperature will be imposed.

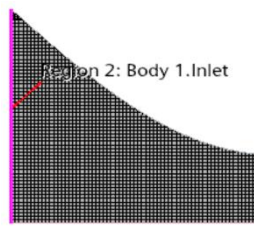


Figure 2 – Inlet region

The values are extracted from the assessment subject, and read:

$$\begin{aligned} - T_{in}^T &= 500 \text{ K}, \\ - P_{in}^T &= 10 \text{ bar}. \end{aligned}$$

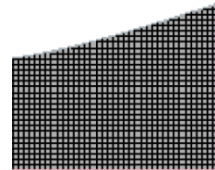


Figure 3 – Outlet region

This region, in an ideal adapted nozzle, brings the fluid at ambient conditions.

We use the Q2 results to evaluate our outlet region quantities.

As tasked by the assessment subject, here are the different visualizations of our framework:

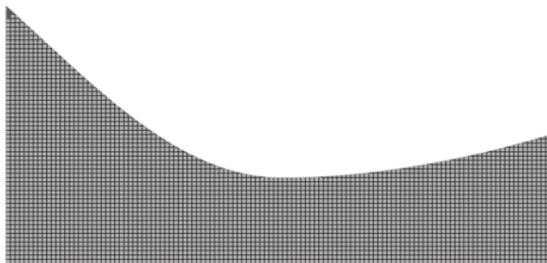


Figure 4 – Mesh Scene

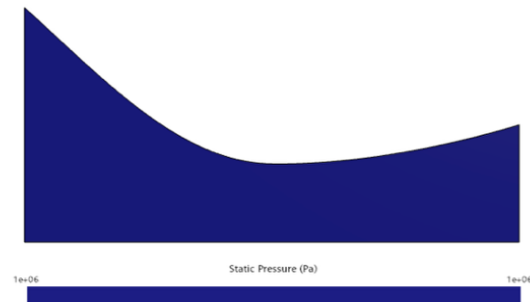


Figure 5 – Initial Pressure Field

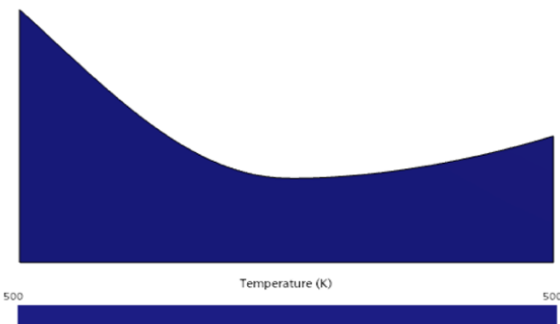


Figure 6 – Initial Temperature Field

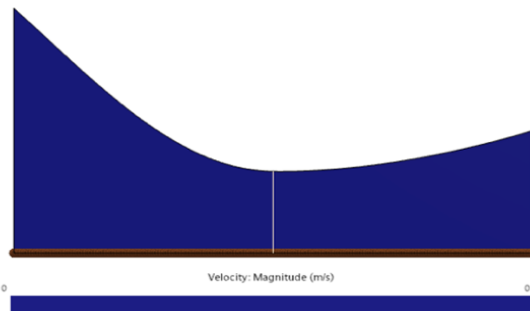
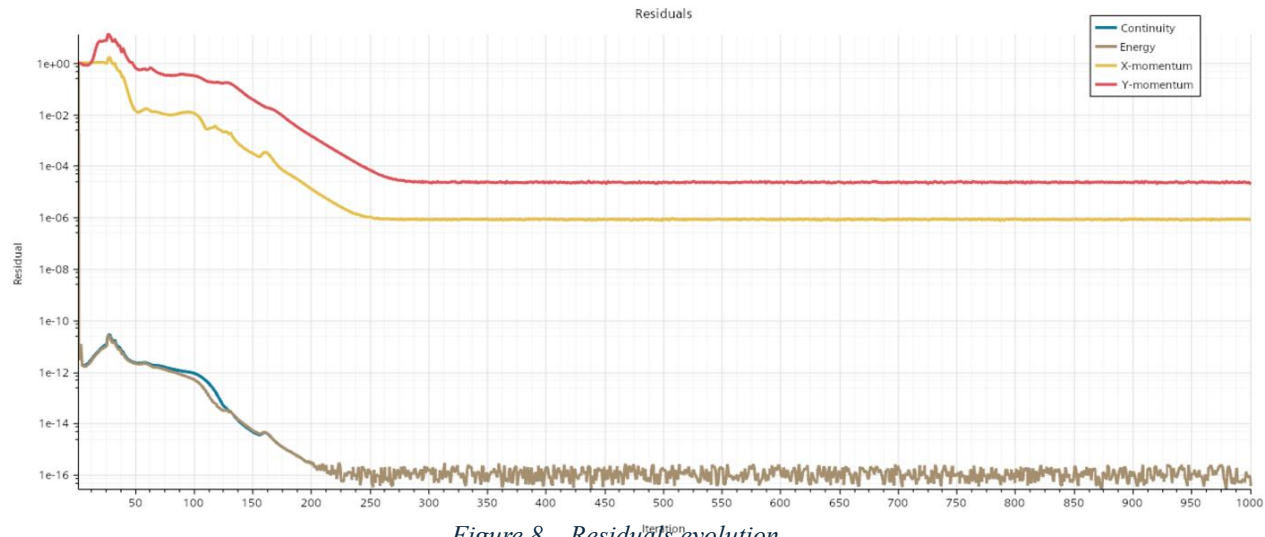


Figure 7 – Initial Velocity Magnitude Field

We, as expected, can observe a homogeneous flow parametrization throughout the nozzle in the first iteration of our resolution.

Question 4 – Converged Simulation Results & Analysis

Below is plotted the most crucial result of our finished simulation, the residual residuals order of magnitude evolution plot. This figure below will confirm or not whether the scalar scenes are even valid to analyze, depending on the residuals behavior indicating a convergence or not.



We observe a **convergence** of the results around **250 iterations**. This assumption is backed up by the stagnation of the residuals after a multiple order of magnitude decrease since iteration 1, a practical convergence criterion for discrete multi-cell resolutions.

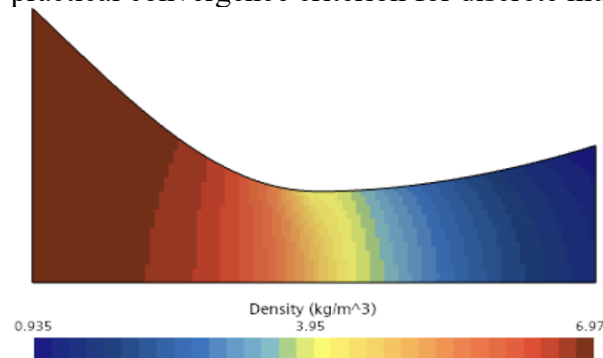


Figure 9 – Converged Density Field

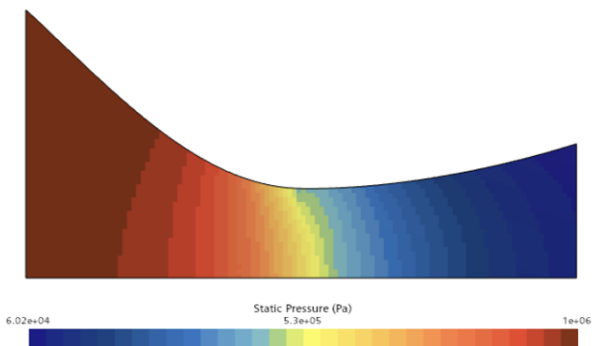


Figure 10 – Converged Pressure Field

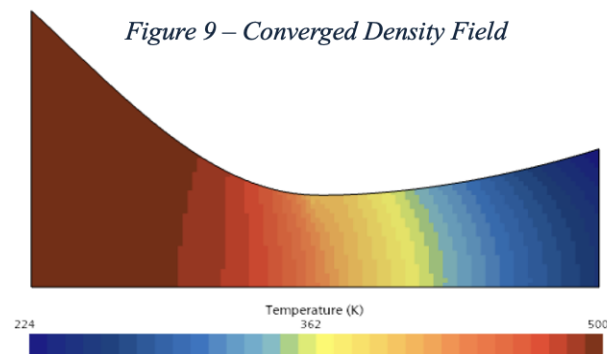


Figure 11 – Converged Temperature Field

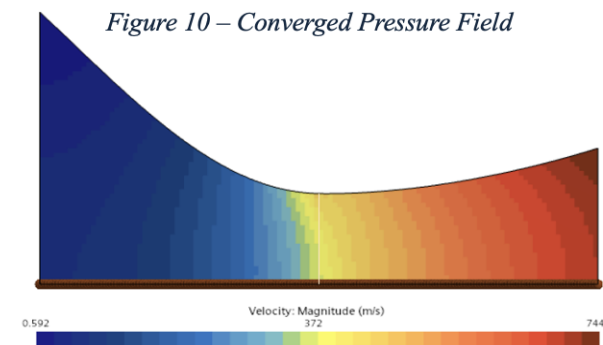


Figure 12 – Converged Velocity Field

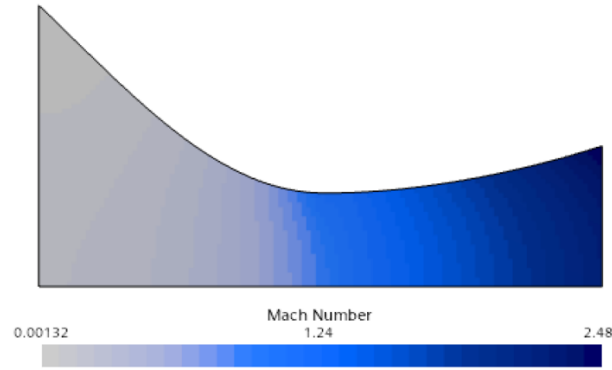


Figure 13 – Converged Mach Field

As expected, we observe a steady **isentropic** (no normal shock) flow of the fluid through the nozzle, with a subsonic – supersonic regime change appearing to occur at the nozzle throat and normal quantities evolution (Hugoniot relation).

Therefore, we seem to have matched with the expected theoretical flow behavior of an **ideally expanded nozzle**.

Question 5 – Theoretical / Numerical $\dot{m}_s$ Comparison.

We can now extract our first numerical value, a central quantity to define a flow in a Laval nozzle: the mass flow rate [kg/s]. Recalling that its value is constant, we extract this value with a Report averaging the mass flow rate at the Outlet section and obtain : 9.003558E-2.

Furthermore, as our model only represents an infinitesimal fraction of the nozzle section, the value given is actually in kg/s/rad. Therefore, multiplying it by 2π rad gives us: $\dot{m}_{s-num} = 0.565 \text{ kg/s}$.

On the other hand, the theoretical value of this mass flow rate is calculated by the below expression (recall that total quantities are conserved from inlet to outlet):

$$\dot{m}_{s-th} = P^T A \sqrt{\frac{\gamma}{r T^T}} M \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{-(\gamma+1)}{2(\gamma-1)}} = 10e5 \cdot 7.068e-4 \sqrt{\frac{1.4}{287 \cdot 500}} 2.33 \left(1 + \frac{1.4-1}{2} 2.33^2 \right)^{\frac{-(1.4+1)}{2(1.4-1)}} = 0.567 \text{ kg/s}$$

With a relative error of: $\frac{0.567-0.565}{0.567} * 100 = 0.35\%$ we can confirm that our simulation **accurately models the flow behavior and quantities** inside our nozzle.

Question 6 – Theoretical / Numerical P_c & out , T_c & out , u_{out} Comparison.

Knowing our theoretical outlet Mach value and throat Mach definition value of 1, we can calculate the static pressure and temperature quantities with the St-Venant equations (ie. [Question 2](#)).

The velocity is calculated from the Mach – air speed expression $M_{out} \sqrt{\gamma r T_{t-out}}$.

Quantity	Theoretical	Numerical	Relative Error [%]
P_c [bar]	5.28	5.44	3.03
T_c [K]	417	420	0.72
P_{out} [bar]	0.765	0.791	3.40
T_{out} [K]	240	242	0.83
u_{out} [m/s]	723	720	0.41

Table 1 – Numerical vs Theoretical comparison of the throat & outlet quantities

We observe global $< 4\%$ relative errors with some quantities reaching $< 1\%$ relative error, confirming the success of our simulation in converging towards a near-theoretical result.

Question 7 – Theoretical / Numerical Profiles Comparison (with num. M).

Finally, exporting the numerical X-Y profiles of static pressure, temperature, and Mach number throughout the nozzle and comparing them with the theoretical (St-Venant) profiles of static pressure and temperature (with the numerical M) gives us the following :

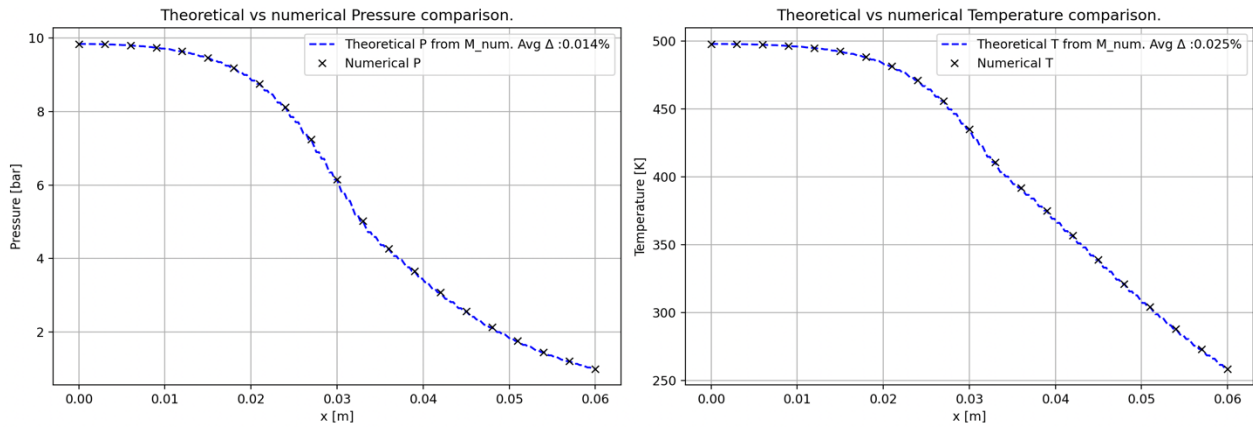


Figure 14 – Numerical static P & T comparison with theoretical values (from num. M)

Once again validating the coherence and success of this first numerical simulation.

II – OVER-EXPANDED NOZZLE

Question 1 – Simulation Initialization

As tasked by the assessment subject, here are the different visualizations of our framework.

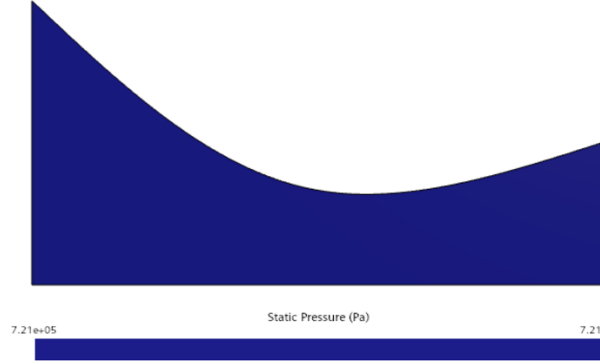


Figure 1 – Initial Pressure Field

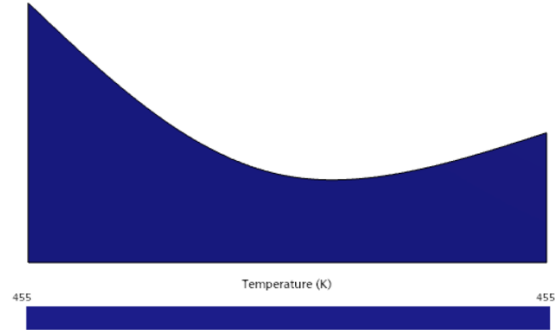


Figure 2 – Initial Temperature Field

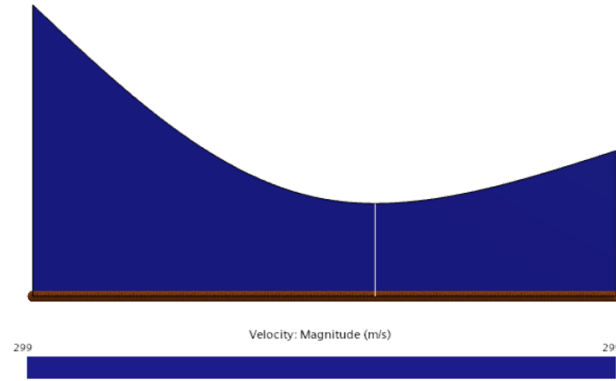


Figure 3 – Initial Velocity Field

As expected, we can observe a homogeneous flow parametrization throughout the nozzle in the first iteration of our resolution.

Furthermore, the thermodynamic consistency of these static quantities guarantee a proper convergence optimization starting framework.

The initial velocity value here is far greater than the previous subject, but subsonic. We arbitrarily choose **Mach 0.7**, implying the following total static initial quantities which, lets recall, match the inlet total quantities:

$$\left\{ \begin{array}{l} P_{s-ini} = P_{t-ini} \left(1 + \frac{\gamma-1}{2} M_{ini}^2 \right)^{-\frac{\gamma}{\gamma-1}} = 10 \left(1 + \frac{1.4-1}{2} 0.7^2 \right)^{-\frac{1.4}{1.4-1}} = 7.21 \Rightarrow \boxed{P_{s-ini} = 7.21 \text{ bar}} \\ T_{s-ini} = T_{t-ini} \left(1 + \frac{\gamma-1}{2} M_{ini}^2 \right)^{-1} = 500 \left(1 + \frac{1.4-1}{2} 0.7^2 \right)^{-1} = 455 \Rightarrow \boxed{T_{s-ini} = 455 \text{ K}} \\ ||u_{ini}|| = M_{ini} \sqrt{\gamma r T_{s-ini}} = 0.7 \sqrt{1.4 \cdot 287 \cdot 455} = 299.4 \Rightarrow \boxed{||u_{ini}|| = 299.4 \text{ m/s}} \end{array} \right.$$

We note that despite the calculations being presented above, the arbitrary choice of the initial Mach 0.7 number can be explained by its large yet still subsonic velocity regime, as well as its order of magnitude closeness with the expected post-shock regime (grossly estimated between 0.5 and 0.9).

Question 2, 3 & 4 – Converged Simulation Results & Analysis

Once again, below is plotted the residuals magnitude evolution throughout the simulation, which should tend towards a convergence if we want to estimate our simulation scalar values as reliable.

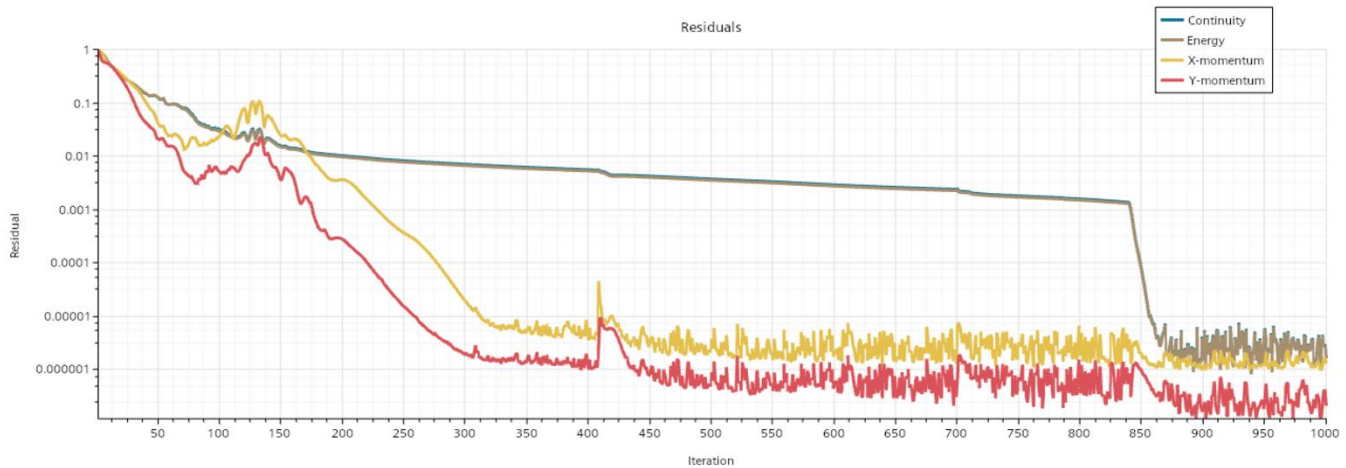


Figure 4 – Residuals Evolution

We observe a final convergence reached after around **870 iterations**, demonstrating a much **less stable numerical framework** to achieve simulation “success” with a steady final form. Nonetheless, our simulation still converged, and we can now observe the scalar fields below:

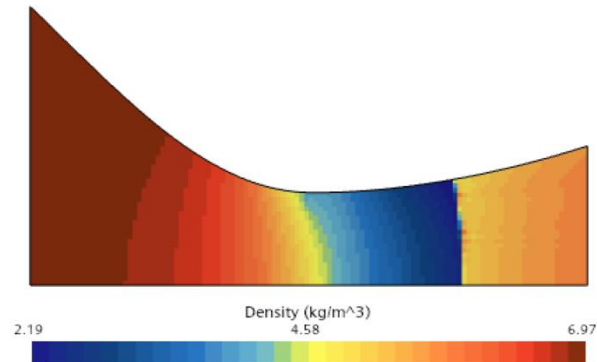


Figure 5 – Converged Density Field

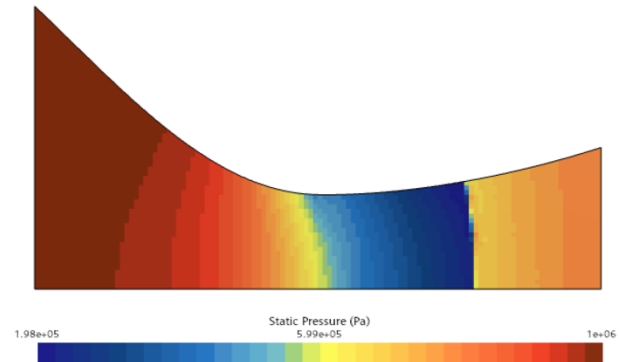


Figure 6 – Converged Pressure Field

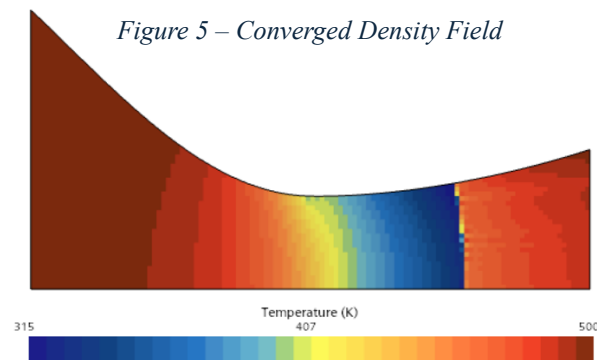


Figure 7 – Converged Temperature Field

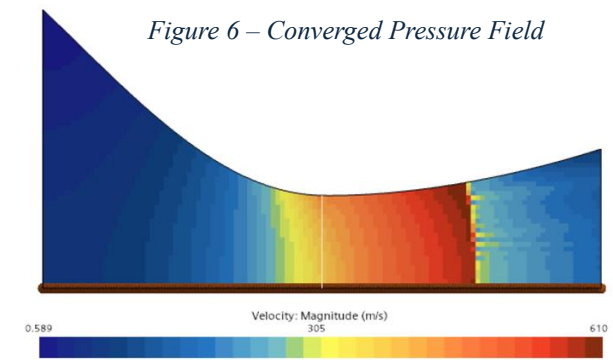


Figure 8 – Converged Velocity Field

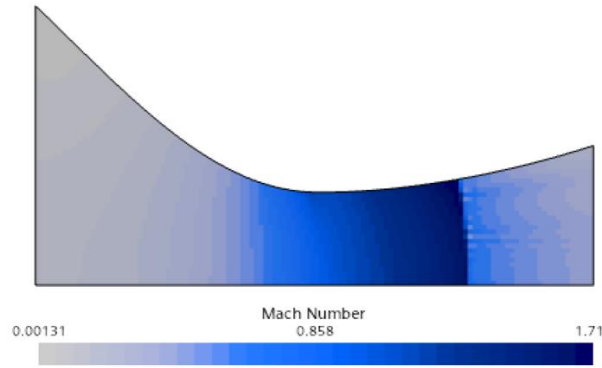


Figure 9 – Converged Mach Field

As expected, we now observe a steady **non-isentropic** (normal shock) flow of the fluid through the nozzle, with a subsonic – supersonic regime change appearing to occur at the nozzle throat before abruptly returning to a subsonic regime downstream of the “normal” shock.

Therefore, we seem to have matched with the expected theoretical flow behavior of an **over-expanded nozzle**.

Note that, especially downstream of the shock, the numerical scalar fields show much less smooth results, indicating once again that the **resolution** of this model represented a **less direct and precise** convergence for a less uniform final solution. Finally, we can also note that the shock occurred well past supersonic regime, respecting thermodynamics second law.

We can now observe the X-Y plots of the pressure, temperature and Mach number quantities throughout the nozzle. Mainly interested in the general behaviors (rather than precise values) linked to this flow, we will extract the Y-values from the bottom axis (center of the nozzle) for simplicity.

We have the following results:

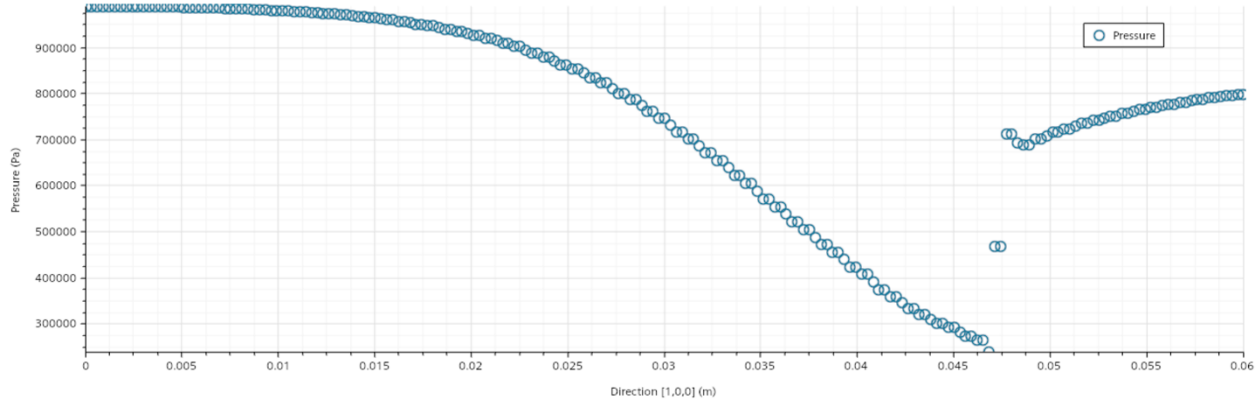


Figure 10 – Converged Pressure profile

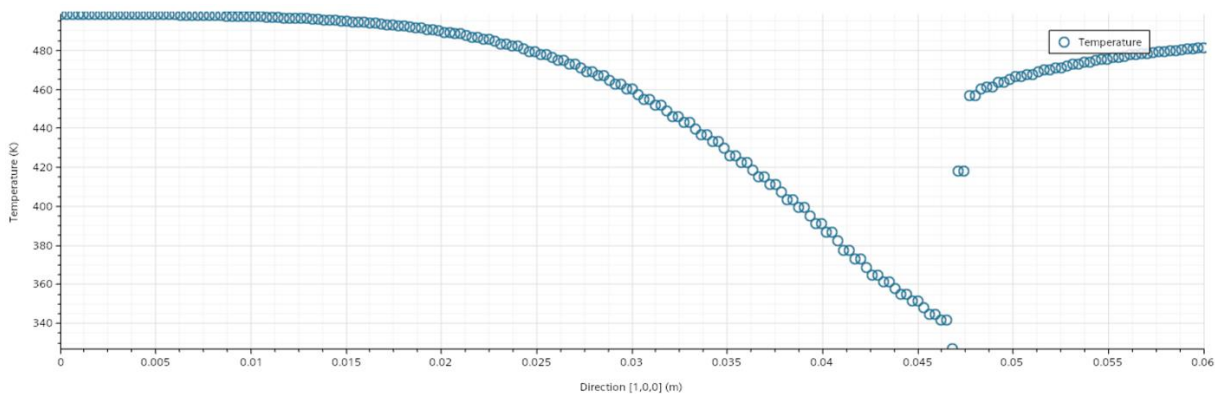


Figure 11 – Converged Temperature profile

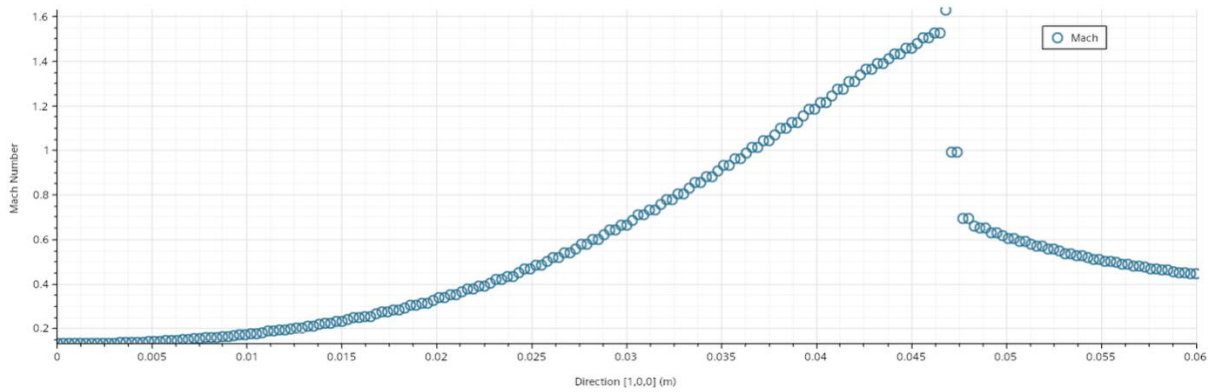


Figure 12 – Converged Mach profile

The results **reflect lecture 3 theoretical behaviors** of the evolution of quantities through a shock jump, with accurate confirmation that the shock occurs at $M=1.5>1$ in the diverging section.

Furthermore, although not displayed on this report to minimize the size of this synthesis of asked results, the plotting of the entropy confirms the fundamental thermodynamical behavior of an overexpanded nozzle: abrupt magnitude jump at the shock longitudinal position.

Question 5 – Mass Flow Rate

The numerical report outputs a value of $\dot{m}_{s-num}' = 0.518 \text{ kg/s}$ equal to the previous subject value before geometrical correction (recall subject 2, geometry does not impose the throat at 0.03m). Thus, we can confirm the fundamental theoretical property stating the conservation of mass flow rate in a nozzle despite the presence of a normal shock or not.

We can compare its new theoretical value by evaluating the new area. To do so, we isolate in the sketch (or by plotting the points composing the top spline) the point where the spline reaches its vertical minimum and identify the new throat upper point coordinate: (0.0352 ; 0.0096).

We therefore solve the same equations as before but with a new throat area : $A_c' = \pi \cdot 0.0096^2$. The result obtained is a new theoretical outlet Mach of $M_{out}' = 2.417$ and a new theoretical mass flow rate of $\dot{m}_{s-th}' = 0.523 \text{ kg/s}$. The $\frac{0.523-0.518}{0.523} * 100 = 0.95\%$ discrepancy remains a high confidence indicator on the coherence of our converged simulation.

Question 6 – Theoretical / Numerical Profiles Comparison (“Laval Shock.py”)

Finally, exporting the numerical X-Y profiles of static pressure and temperature throughout the nozzle and comparing them with the theoretical (Mach and quantities jump relations) profiles of static pressure and temperature (with Laval Shock.py) gives us the following :

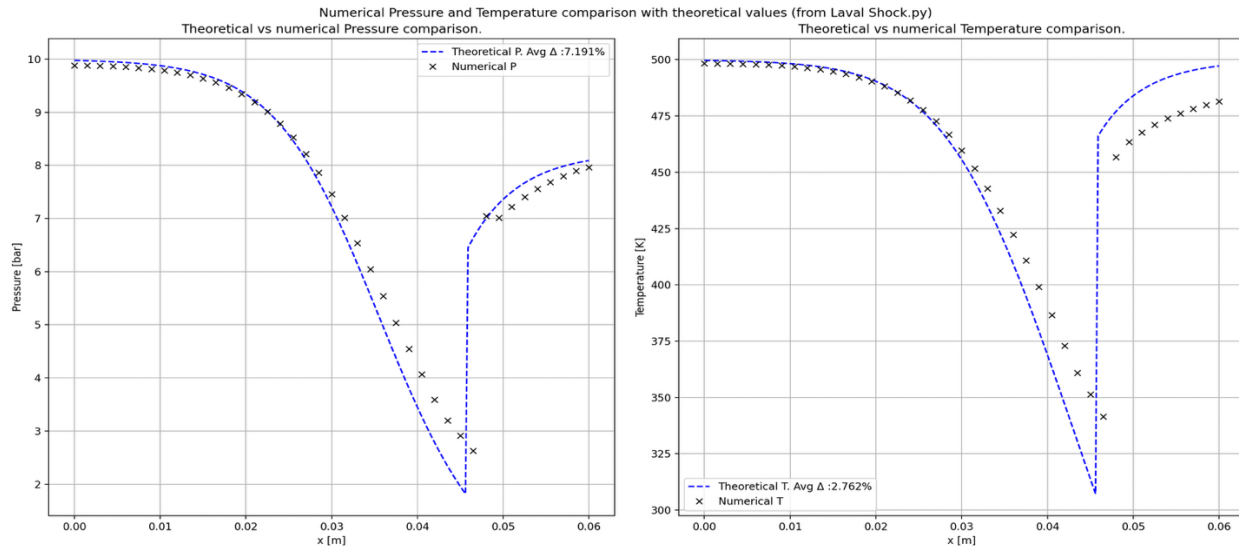


Figure 13 – Numerical static P & T comparison with theoretical values

Once again validating the coherence and success of this second numerical simulation, despite a slightly higher average relative error from the less numerically stable model framework.

Lastly, we can compare the observed shock Mach of 1.626 to the theoretical one of 1.772 by the code to confirm once again the relevancy of our simulation results. Relative error : 8.2%.

III – OVER-EXPANDED JET

Question 1 – Simulation Initialization

The non-free jet framework with the coflow parameters, as well as the initial and boundary conditions follow the assignment guidelines. Furthermore, we maintain a **reference pressure of 0 Pa** (working with relative pressures).

The numerical framework of this jet plume simulation being much more unstable than the previous ones, we initialize this simulation using the Grid Sequencing algorithm: firstly evaluating the simulation on a coarse grid, before slowly refining it to obtain a near-end state initial solution.

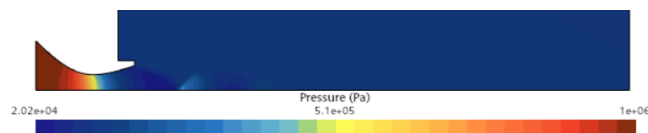


Figure 1 – Initial pressure field

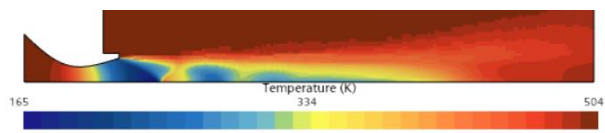


Figure 2 – Initial temperature field

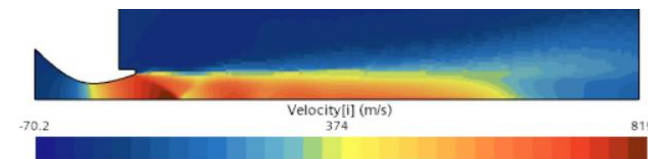


Figure 3 – Initial velocity field

As expected, we observe very different fields from the previous tutorials where homogeneous initial conditions were used.

In this case, a plume-like jet profile appears after the GS calculation, very much helping our later simulation to converge easier and avoid divergence.

Question 2 – Converged Simulation Results

After a much longer simulation, with an increased maximum number of steps to 3000, we obtain the following residuals plot monitoring.

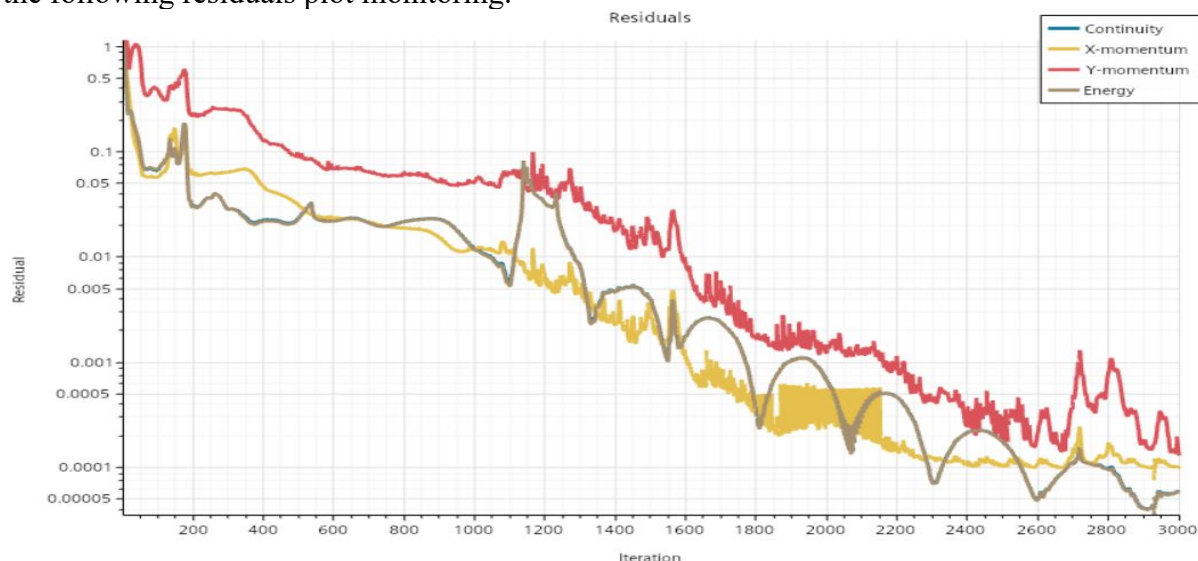


Figure 4 – Residuals plot

And the following thermodynamic, Mach, and numerically approximated Schlieren fields:

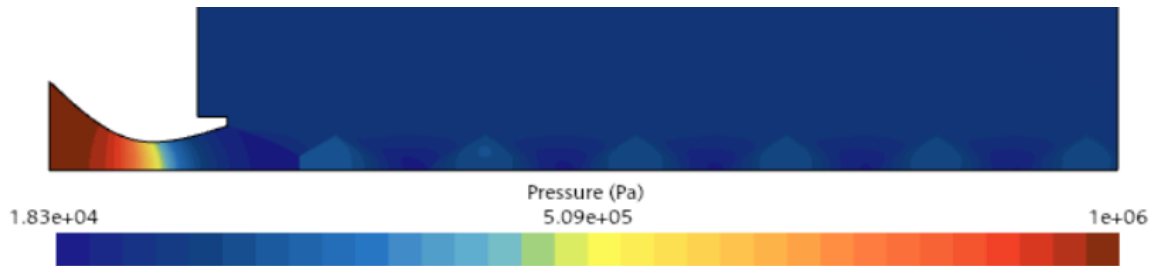


Figure 5 – Converged pressure field

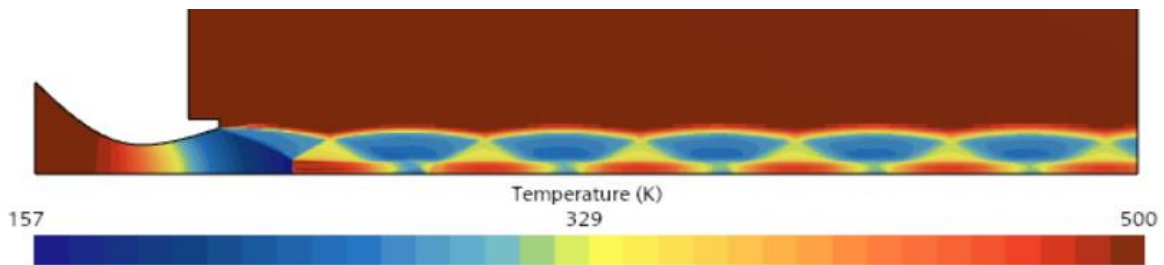


Figure 6 – Converged temperature field

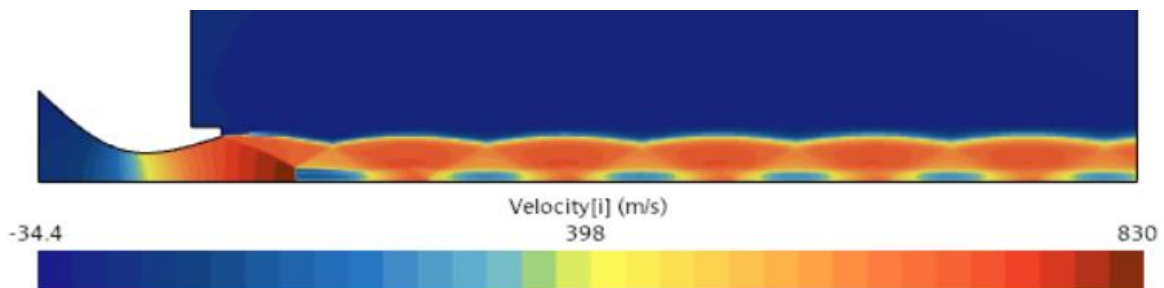


Figure 7 – Converged velocity field



Figure 8 – Converged density field

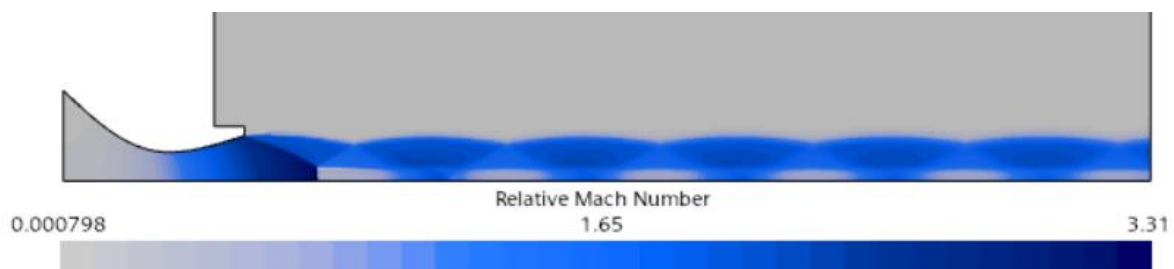


Figure 9 – Converged Mach field

For this study, note that we added the normalized density variation Schlieren scalar, computed by the `mag(grad($Density))/$Density` code inside our newly user-created field function.

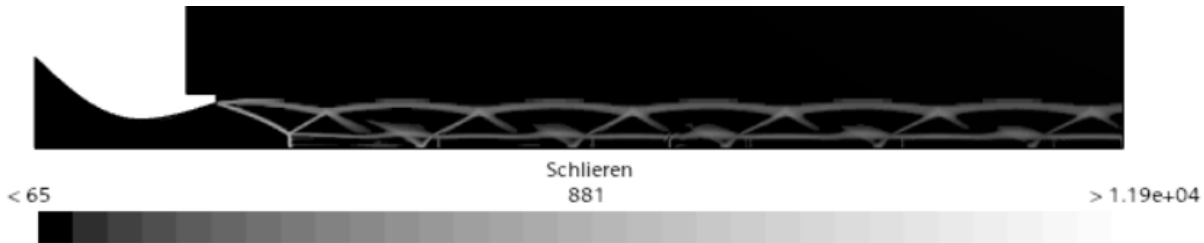


Figure 10 – Converged Schlieren field

We can validate our jet plume simulation, with the jet core structure matching the theoretical over-expanded jet model viewed in our turbomachinery lectures: shock – Mach disk – oblique shock – repeat of these cells. The thermodynamic quantities in the plume evolve in regions, separated by the reflecting shock waves and expansion fans. In the nozzle, they obey similarly to our ideally expanded nozzle configuration.

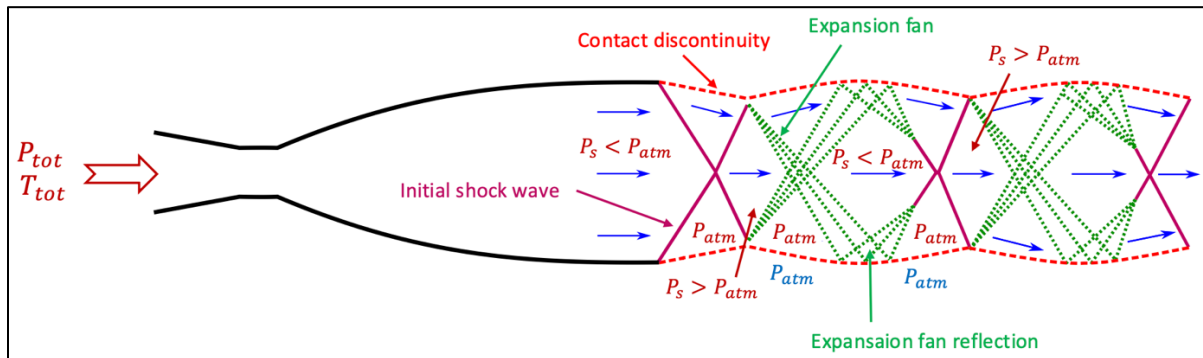


Figure 11 – Theoretical jet structure

Question 3 – Nozzle Quantities Profiles

Still with an over-expanded nozzle but without creation of a normal shock, we can now compare the nozzle profiles of the Pressure, Temperature and Mach number:

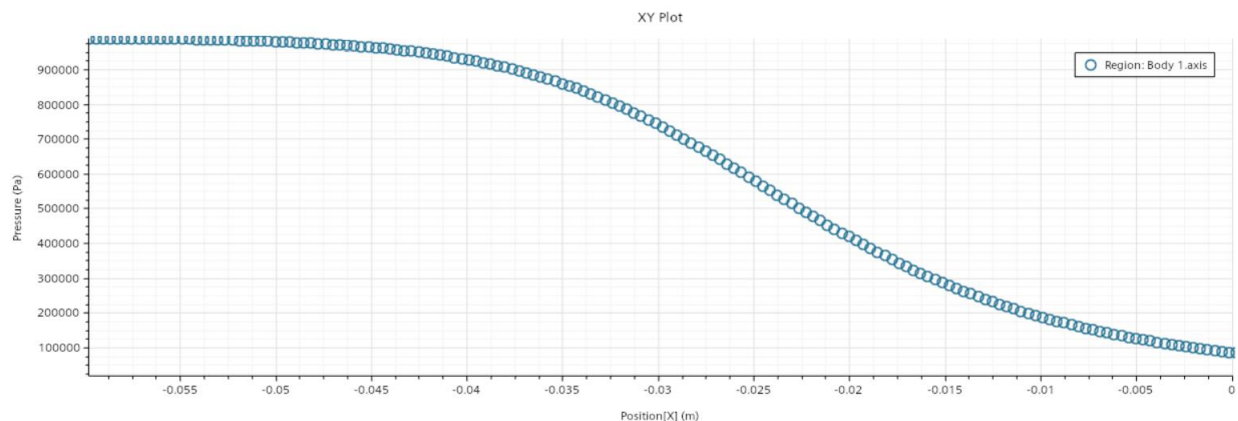


Figure 12 – Nozzle pressure profile

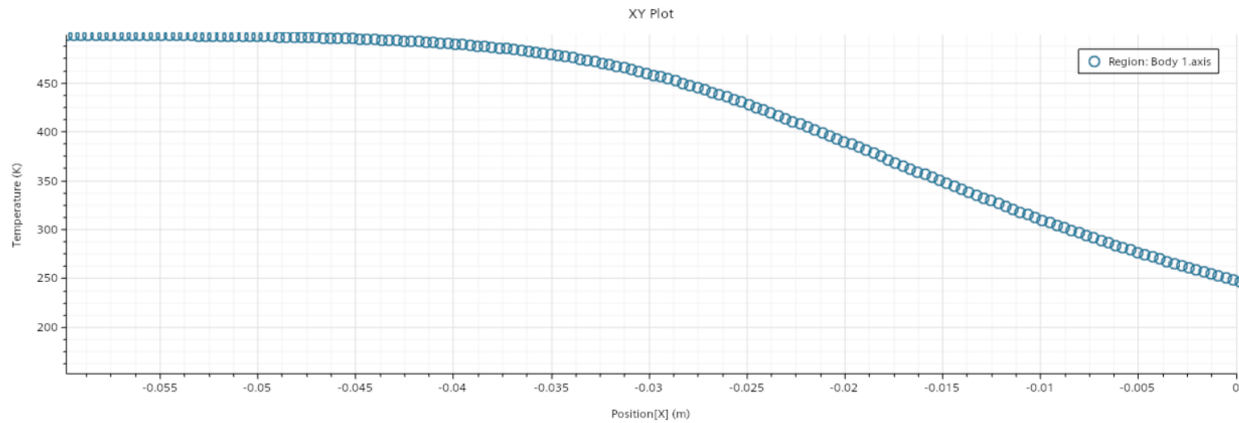


Figure 13 – Nozzle temperature profile

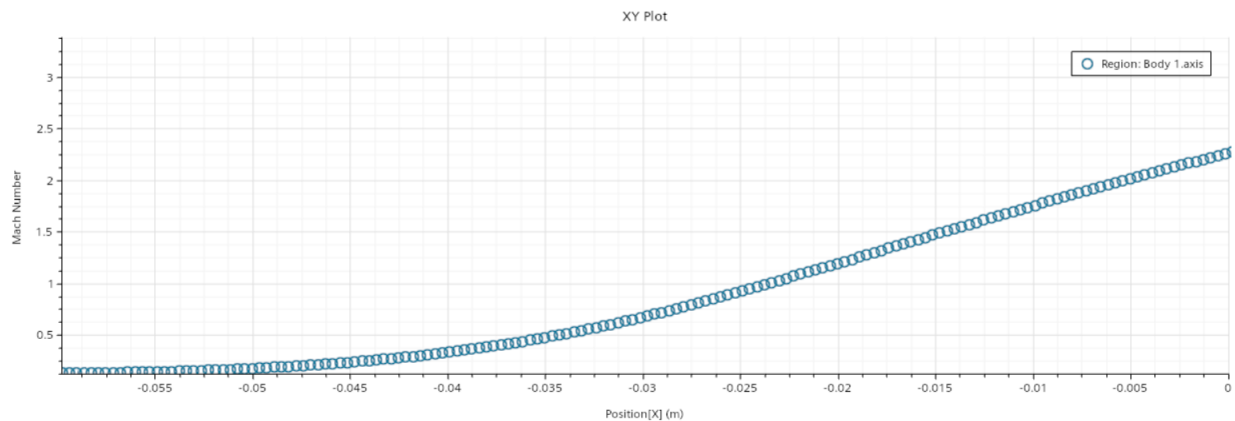


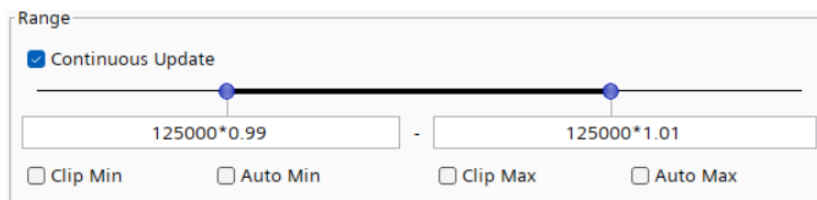
Figure 14 – Nozzle Mach profile

Note that the x-axis is counted negative due to the origin being here imposed at the nozzle outlet.

Question 4 – Effective / Target Ambient Pressure Comparison

As the true free jet framework poses many numerical instability issues, recall that we impose a low velocity coflow for the ambient field. This method therefore leads to a simulation result where the effective converged ambient pressure is slightly different than a target ambient pressure of 1.25bar.

To ensure this error is negligible in our result, we can extract the effective ambient pressure in our scalar pressure scene and verify it fits in the $1.25\text{bar} \pm 1\%$ = $[1.2375 : 1.2625]$ bar interval.



To do so, we shrink our color bar min. and max. values accordingly as shown opposite.

We can now visually confirm that the ambient pressure not only fits inside this range but also matches the near-middle of it as shown below.

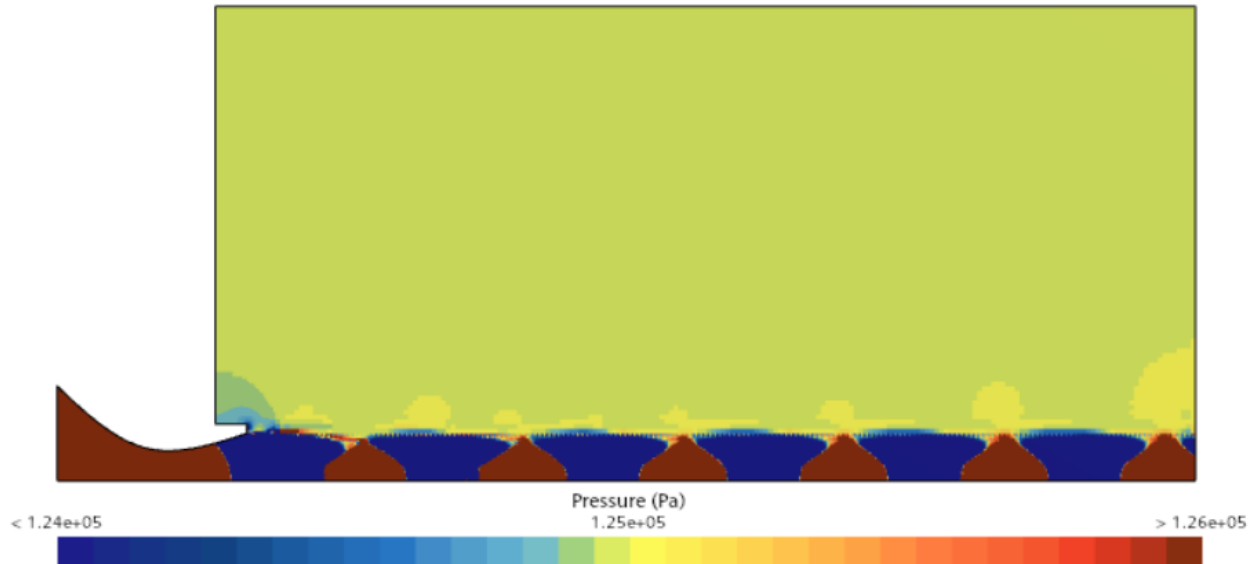


Figure 15 – Re-scaled pressure field

As we can see, the ambient pressure visually fits the error interval and values itself near 1.25bar.

Question 5 – Lip Shock Theoretical Pressure Jump & Mach Angle

Knowing the theoretical pressure jump equation downstream (1) and upstream (0) of the shock -

$$\frac{P_1}{P_0} = 1 + \frac{2\gamma}{\gamma + 1} (M_{n0}^2 - 1)$$

- we can isolate the upstream Mach number M_{n0} and therefore Mach angle σ of the first upper nozzle lip shock. Indeed, since the flow before the outlet shock is ideally expanded, the resulting theoretical outlet pressure and Mach values are the same as the first simulation ones.

Quantity (theoretical)	Value
P_0 [bar]	0.765
P_1 [bar]	1.25
M_0 [Ø]	2.33

Table 1 – Theoretical values at lip shock

We can now establish the expression and calculate the value of the Mach angle :

$$\sigma = \arcsin \left(\frac{1}{M_o} \sqrt{\left(\frac{P_1}{P_0} - 1 \right) \frac{\gamma + 1}{2\gamma} + 1} \right) = \arcsin \left(\frac{1}{2.33} \sqrt{\left(\frac{1.25}{0.765} - 1 \right) \frac{1.4 + 1}{2 \cdot 1.4} + 1} \right)$$

And obtain :

Quantity (theoretical)	Value
σ [°]	32.21

Table 2 – Theoretical Mach angle at lip shock

Question 6 – Theoretical / Numerical Mach Angle Comparison

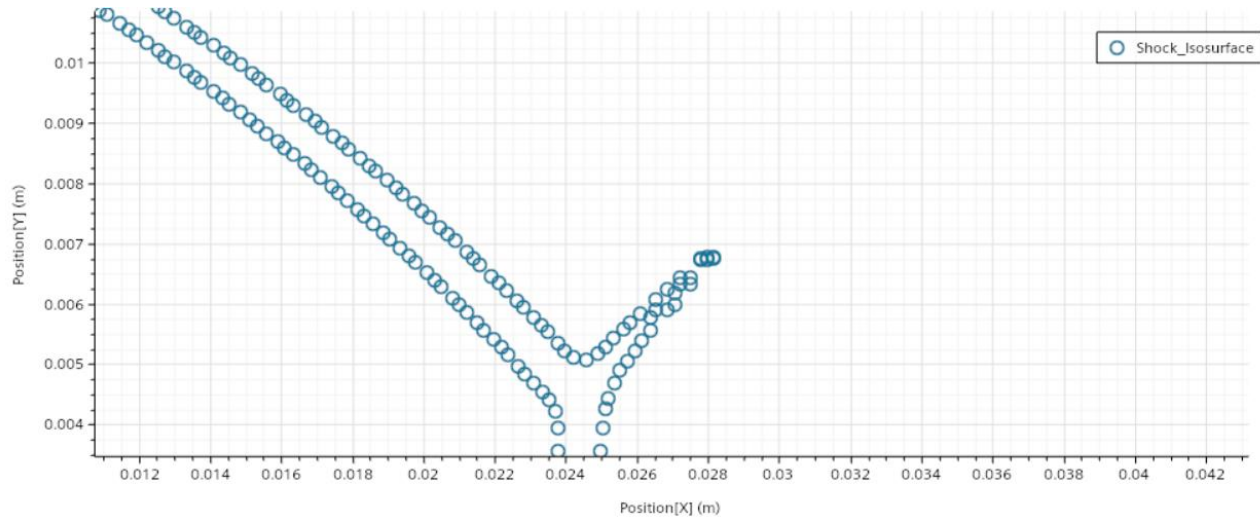


Figure 16 – First plume shock reflection

Following the assignment guidelines, we extract the point: $(x^*, y^*) = (0.022, 0.006)$.

Thanks to the shown below geometrical expression and knowing that the outlet nozzle wall forms an $\alpha = 17.8448^\circ$ angle with the horizontal axis, we are able to calculate the numerical Mach angle:

$$\sigma_{num} = \alpha + \arctan\left(\frac{y_{out} - y^*}{x^*}\right) = 17.8448 + \arctan\left(\frac{0.015 - 0.006}{0.022}\right)$$

The calculation therefore gives us:

Quantity	Theoretical	Numerical	Relative Error [%]
σ [°]	32.21	40.09	23.7

Table 3 – Theoretical & Numerical Mach angle comparison

This high relative error value could be expected and can mainly be explained by considerable differences between the theoretical and numerical calculation framework such as :

- The geometrical procedures error in calculating the numerical Mach angle.
- The numerical “realistic” shock wave curved profile, since quantities transported change along the wave. In our theoretical model on the other hand, the shock wave is a straight line, calculated with an un-appropriate angle expression for a proper physical shock wave.

Question 7 – Pressure & Shock Behavior in Different Nozzles

Looking back at our tutorial 1,2 and now 3 results, we can now complete the assignment summary table with **orange answers** to distinguish from the partly completed assignment answers :

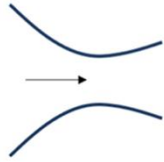
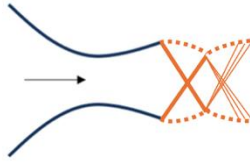
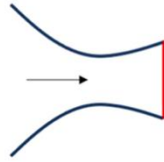
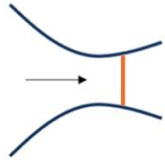
	Ideally expanded nozzle	Over-Expanded jet	Normal shock at nozzle outlet	Over-Expanded nozzle
P_{∞}	0.765 bar	1.25 bar	4.72 bar *	8 bar
Flow structure				

Table 4 – Overall study flow properties comparison

* calculated from the pressure jump equation (derived from the momentum conservation equation) applied at the ideally expanded nozzle outlet values. We have therefore :

$$\frac{p_1}{p_0} = 1 + \frac{2\gamma}{\gamma + 1} (M_0^2 - 1) = 1 + \frac{2 \cdot 1.4}{1.4 + 1} (2.33^2 - 1) = 6.17$$

$$\Leftrightarrow p_{\infty} = p_1 = 6.17 \cdot p_0 \Rightarrow \boxed{p_{\infty} = 6.17 \cdot 0.765 = 4.72 \text{ bar}}$$